

```
In [19]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: data = sns.load_dataset('titanic')
```

```
In [3]: data.head()
```

```
Out[3]:
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adu
0	0	3	male	22.0	1	0	7.2500	S	Third	man	
1	1	1	female	38.0	1	0	71.2833	C	First	woman	
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	
3	1	1	female	35.0	1	0	53.1000	S	First	woman	
4	0	3	male	35.0	0	0	8.0500	S	Third	man	

```
In [4]: data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
#   Column          Non-Null Count  Dtype
---  -
0   survived        891 non-null    int64
1   pclass          891 non-null    int64
2   sex             891 non-null    object
3   age            714 non-null    float64
4   sibsp          891 non-null    int64
5   parch          891 non-null    int64
6   fare           891 non-null    float64
7   embarked       889 non-null    object
8   class          891 non-null    category
9   who            891 non-null    object
10  adult_male     891 non-null    bool
11  deck          203 non-null    category
12  embark_town    889 non-null    object
13  alive          891 non-null    object
14  alone         891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB
```

```
In [5]: data.isnull().sum()
```

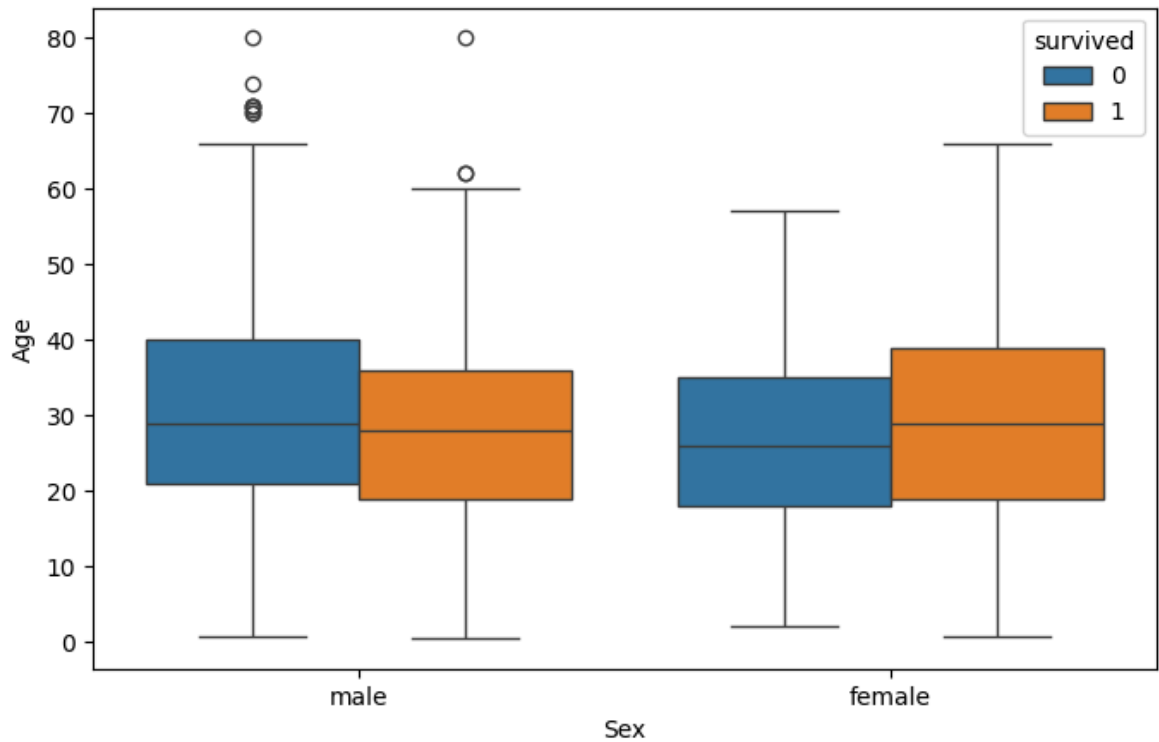
```
Out[5]: survived      0
        pclass        0
        sex           0
        age          177
        sibsp         0
        parch         0
        fare          0
        embarked      2
        class         0
        who           0
        adult_male     0
        deck          688
        embark_town    2
        alive          0
        alone          0
        dtype: int64
```

```
In [6]: data = data.bfill()
```

```
In [7]: data.isnull().sum()
```

```
Out[7]: survived      0
        pclass        0
        sex           0
        age           0
        sibsp         0
        parch         0
        fare          0
        embarked      0
        class         0
        who           0
        adult_male     0
        deck          1
        embark_town    0
        alive          0
        alone          0
        dtype: int64
```

```
In [10]: plt.figure(figsize=(8,5))
        sns.boxplot(x='sex', y='age', hue='survived', data=data)
        plt.xlabel("Sex")
        plt.ylabel("Age")
        plt.show()
```



```
In [12]: mean_age = data['age'].mean()
print(mean_age)
```

29.87056116722783

```
In [14]: std_age = data['age'].std()
print(std_age)
```

14.597667657302386

```
In [16]: data['zscore'] = (data['age'] - mean_age) / std_age
```

```
In [17]: data['zscore']
```

```
Out[17]: 0    -0.539166
         1     0.556900
         2    -0.265149
         3     0.351388
         4     0.351388
         ...
        886  -0.196645
        887  -0.744678
        888  -0.265149
        889  -0.265149
        890   0.145875
         Name: zscore, Length: 891, dtype: float64
```

```
In [20]: outliers = data[np.abs(data['zscore']) > 3]
outliers
```

Out[20]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_
629	0	3	male	80.0	0	0	7.7333	Q	Third	man	
630	1	1	male	80.0	0	0	30.0000	S	First	man	
851	0	3	male	74.0	0	0	7.7750	S	Third	man	



In [21]: `titanic_cleaned = data[np.abs(data['zscore']) <= 3]`

In [22]: `titanic_cleaned = titanic_cleaned.drop(columns=['zscore'])`

In [23]: `print("Original dataset size:", data.shape[0])`
`print("Cleaned dataset size:", titanic_cleaned.shape[0])`

Original dataset size: 891
 Cleaned dataset size: 888

In [24]: `titanic_cleaned.head()`

Out[24]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adu
0	0	3	male	22.0	1	0	7.2500	S	Third	man	
1	1	1	female	38.0	1	0	71.2833	C	First	woman	
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	
3	1	1	female	35.0	1	0	53.1000	S	First	woman	
4	0	3	male	35.0	0	0	8.0500	S	Third	man	



In []: